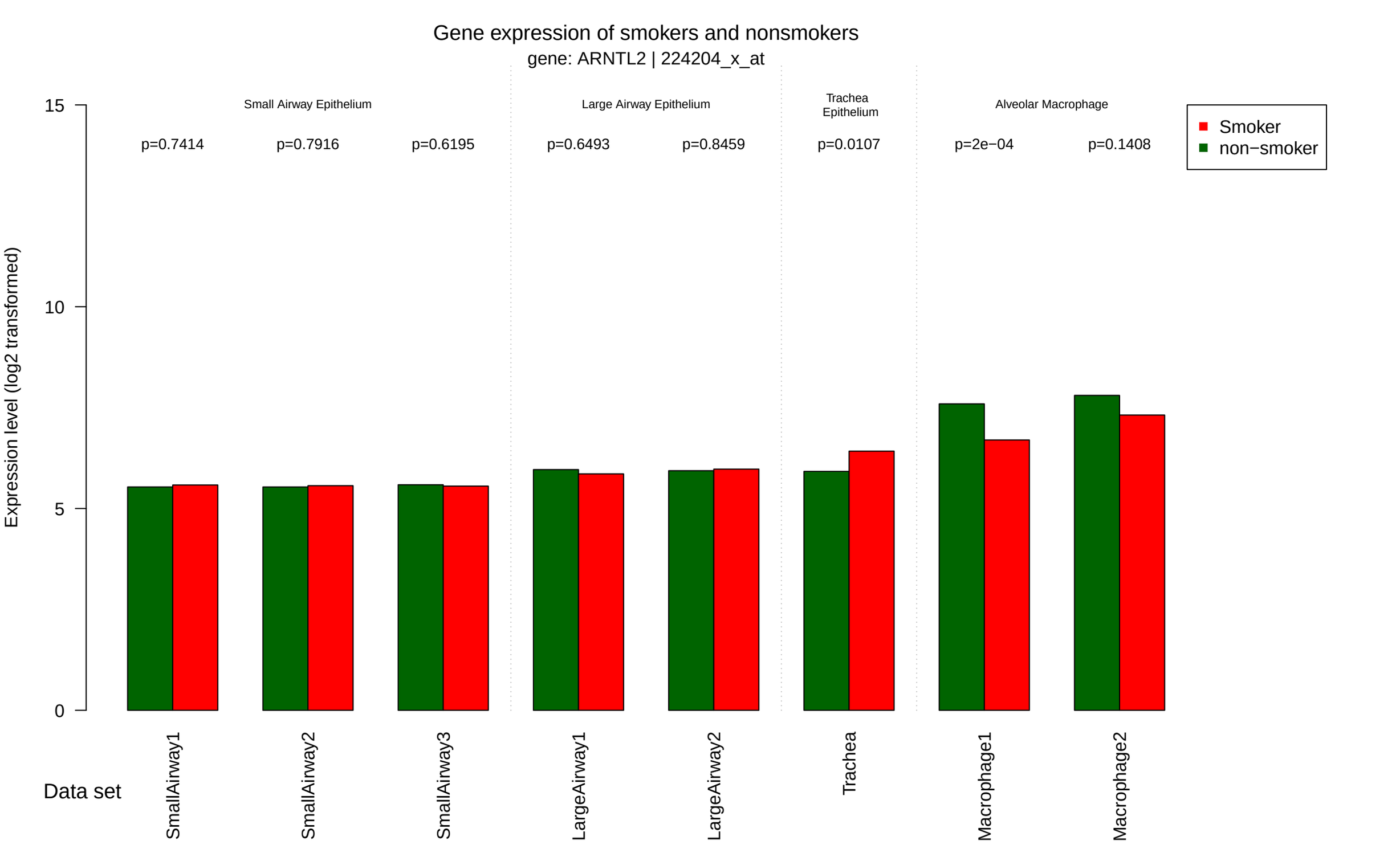
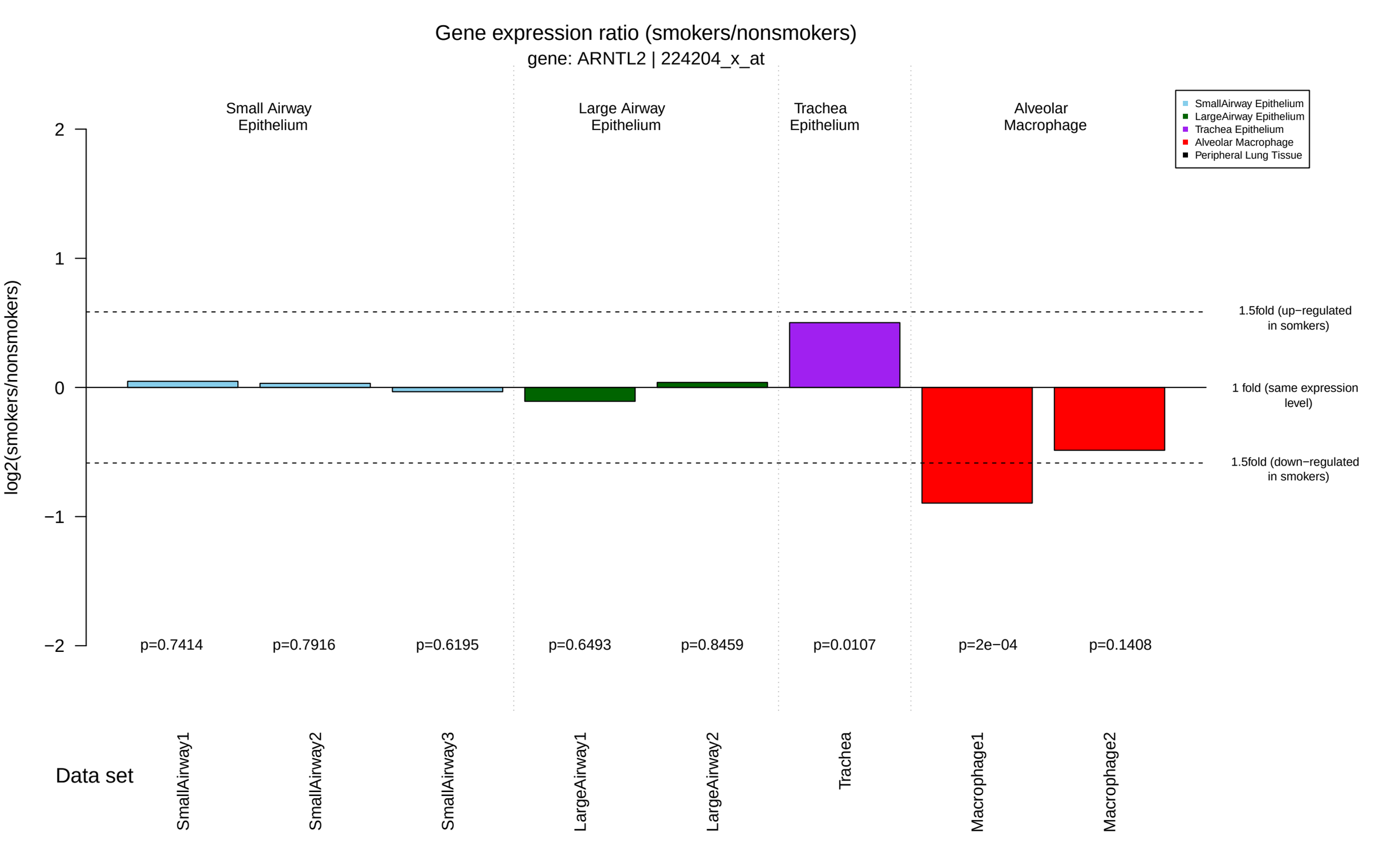
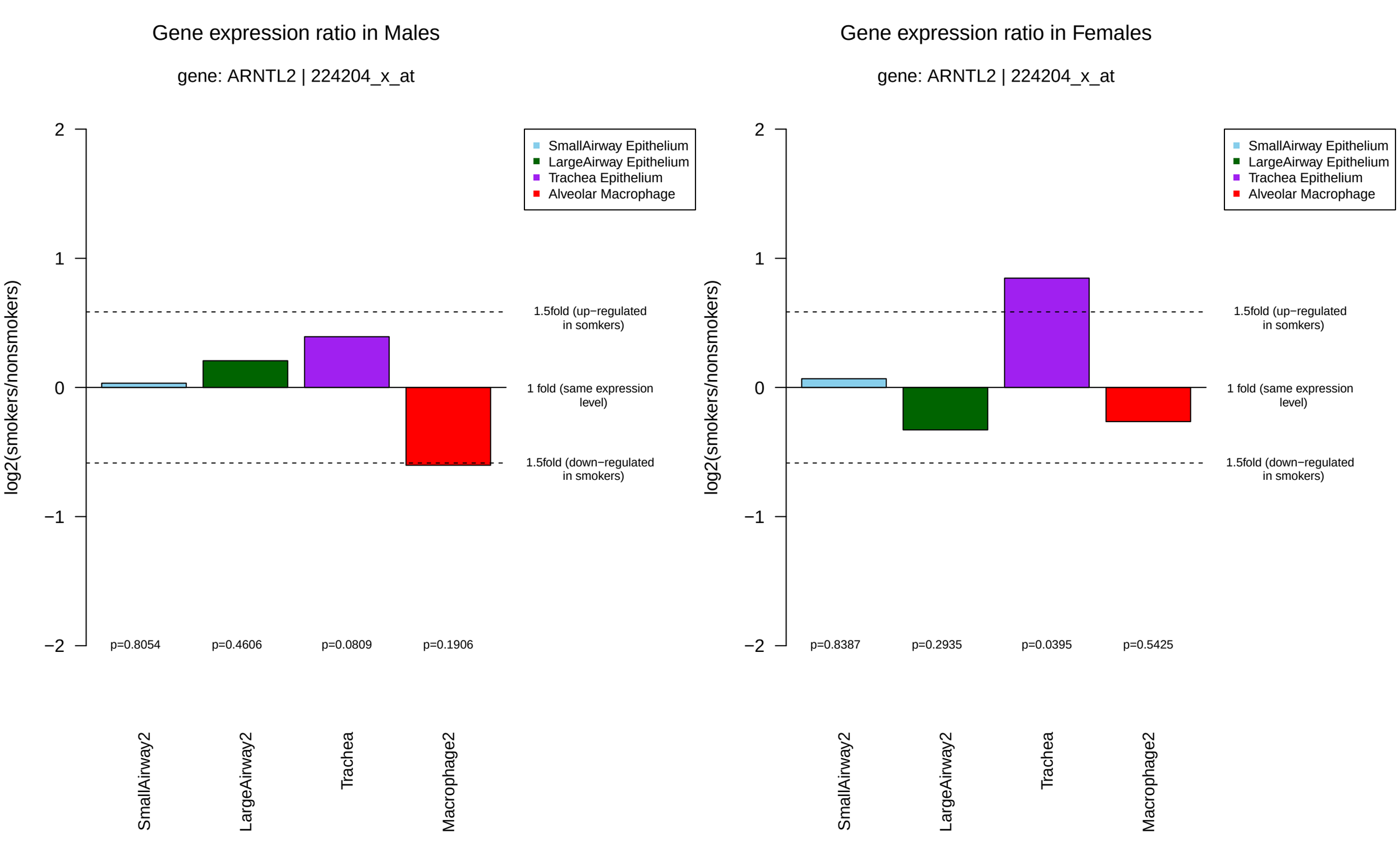
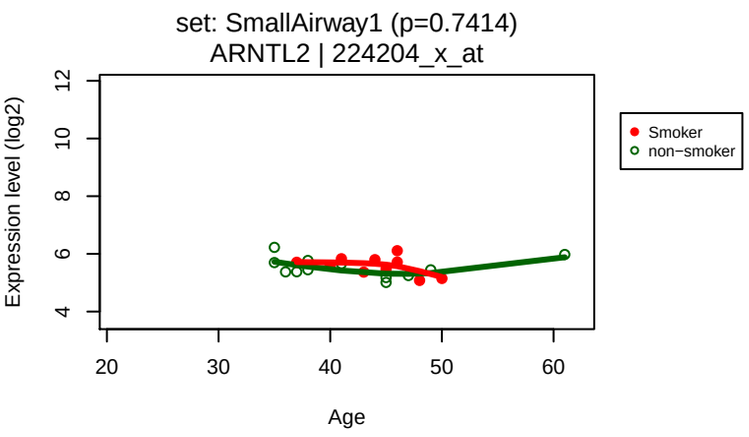
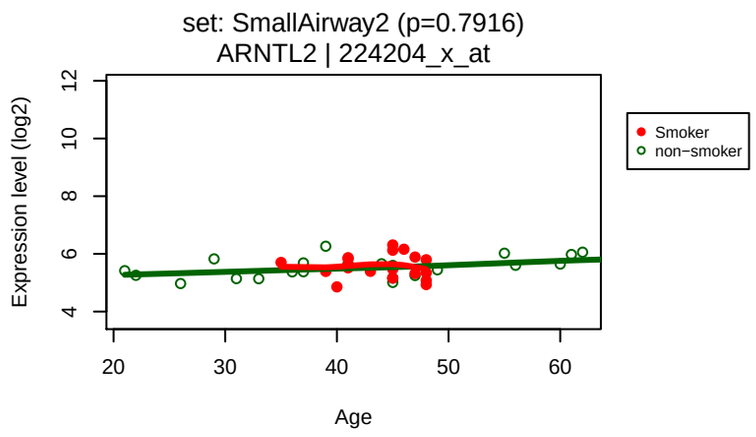
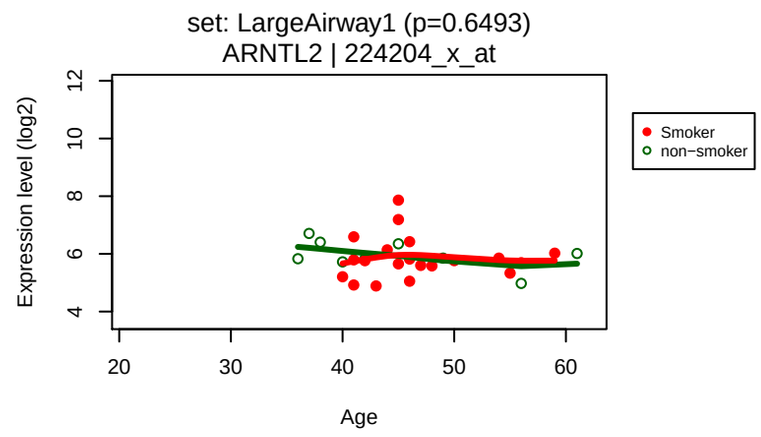

Gene expression of smokers and nonsmokers
gene: ARNTL2 | 224204_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7414
p=0.7916
p=0.6195
p=0.6493
p=0.8459
p=0.0107
p=2e−04
p=0.1408
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

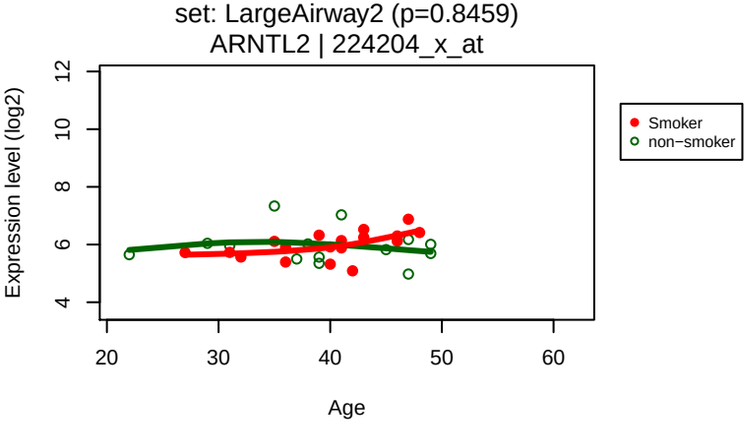

Gene expression ratio (smokers/nonsmokers)
gene: ARNTL2 | 224204_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7414
p=0.7916
p=0.6195
p=0.6493
p=0.8459
p=0.0107
p=2e−04
p=0.1408
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

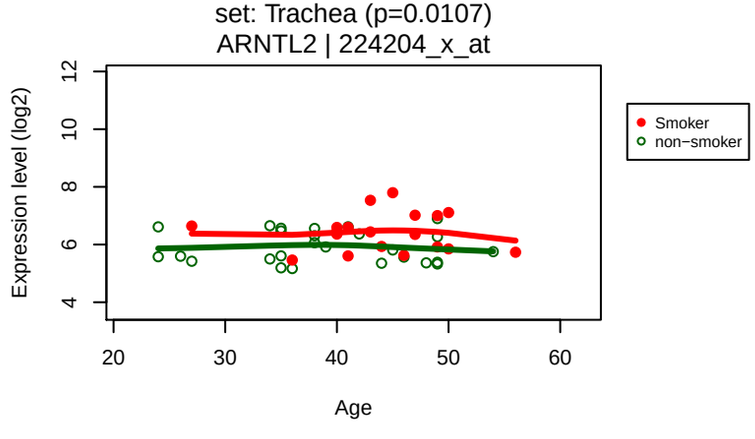

Gene expression ratio in Males
gene: ARNTL2 | 224204_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8054
p=0.4606
p=0.0809
p=0.1906
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARNTL2 | 224204_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8387
p=0.2935
p=0.0395
p=0.5425
SmallAirway2
LargeAirway2
Trachea
Macrophage2

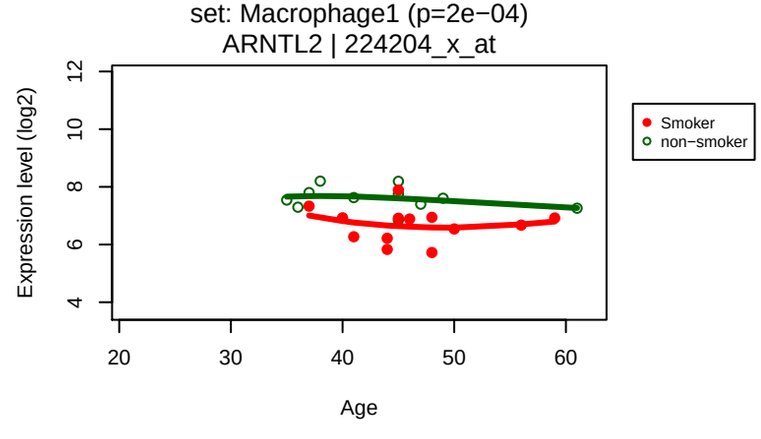

set: SmallAirway1 (p=0.7414)
ARNTL2 | 224204_x_at
Expression level (log2)
Age
Smoker
non−smoker

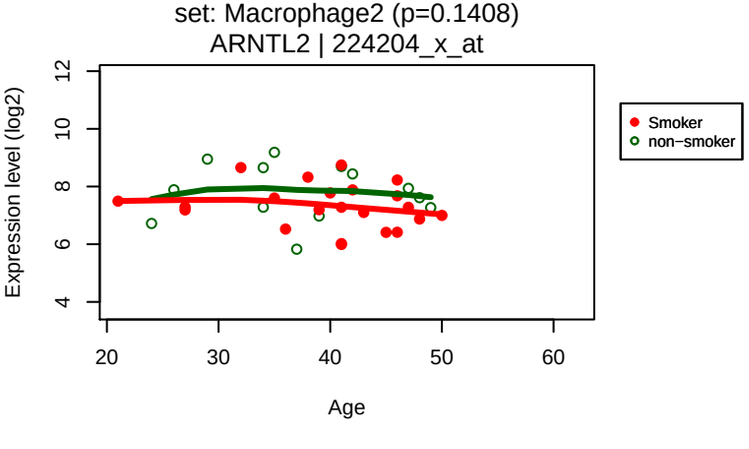

set: SmallAirway2 (p=0.7916)
ARNTL2 | 224204_x_at
Expression level (log2)
Age
Smoker
non−smoker

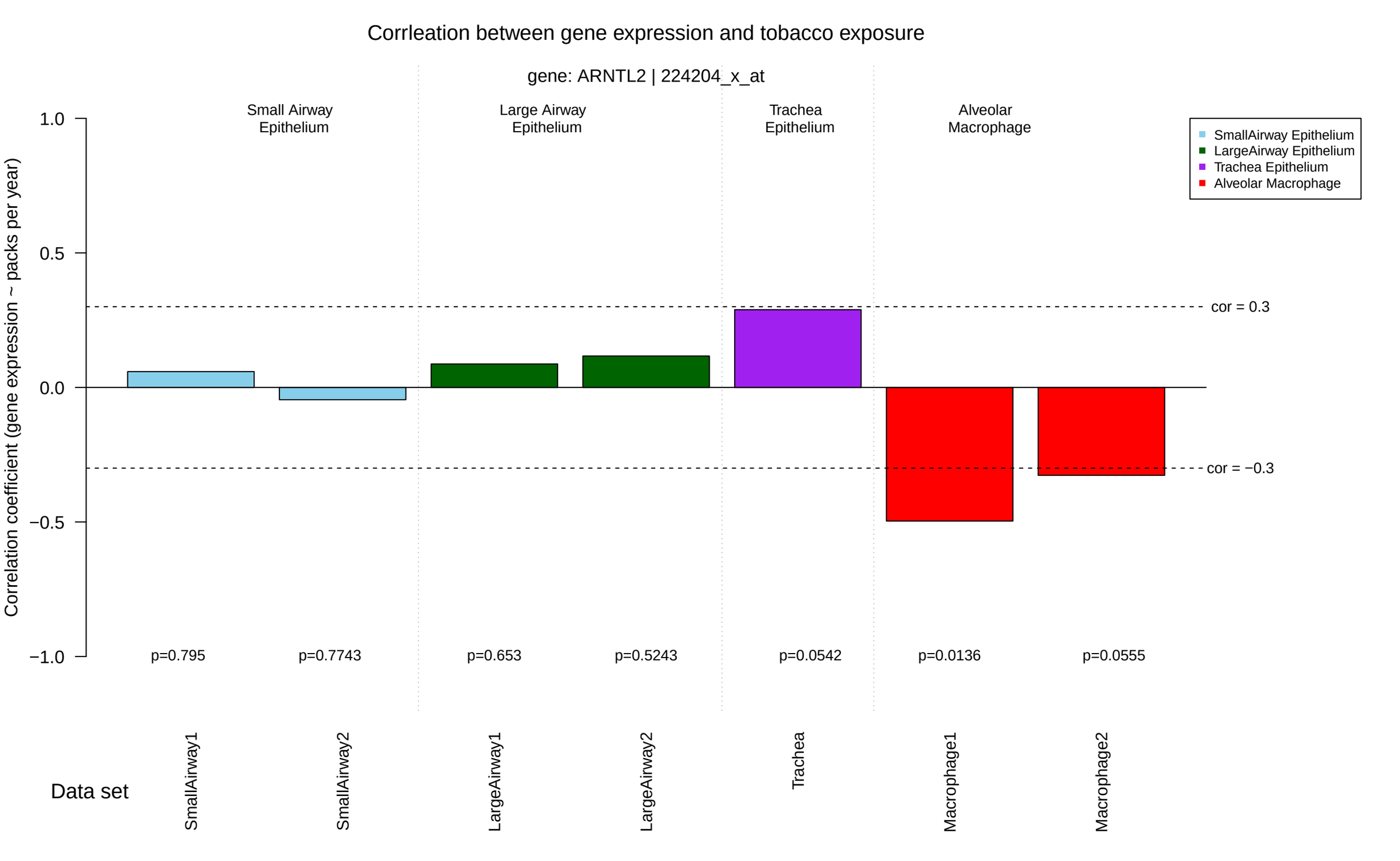

set: LargeAirway1 (p=0.6493)
ARNTL2 | 224204_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8459)
ARNTL2 | 224204_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0107)
ARNTL2 | 224204_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=2e−04)
ARNTL2 | 224204_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1408)
ARNTL2 | 224204_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARNTL2 | 224204_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.795
p=0.7743
p=0.653
p=0.5243
p=0.0542
p=0.0136
p=0.0555
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2